FRED ZHENGYU FU

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EDUCATION

University of Florida — ***Gainesville, United States***

- **Research Master in Forest Resources and Conservation** — *July 2025 –Present*
  Advisor: Dr. Jiangxiao Qiu **|** Current GPA: **4.0/4.0**
- **Non-degree Research Student** — *Sep 2024 - July 2025*
  Research Assistant to Dr. Jiangxiao Qiu **|** GPA: 4.0/4.0

Zhejiang University — ***Hangzhou, China***

- **B.S. in Ecology** — *Sep 2020 - Jun 2025*
- GPA:3.3/4.0 | **junior year GPA 3.83/4.0 (Major Courses)**
- Advisor: Dr. Shuang liang and Dr. Weile Chen

Core Courses

University of Florida

- Analysis of Forest Ecosystem (100/100)
- Forest Ecosystem Health (97/100)
- Image Processing of Remote Sensing (95/100)
- Geospatial Application of UAS (94/100)
- Natural Resources Climate Change (94/100)
- Geospatial AI for Sustainable Science (92/100)

Zhejiang University

- Conservation Biology (94/100)
- Restoration Ecology (92/100)
- Environmental Biology I (93/100)
- Environmental Biology II (92/100)
- Field Ecology (91/100)
- Global Change Ecology (91/100)
- Community and Ecosystem Ecology (91/100)
- Population Ecology (90/100)

RESEARCH EXPERIENCE

University of Florida

Coupling Process-Based Modeling and Machine Learning to Predict Grassland Productivity and Environmental Responses in Florida

Project Leader — *Sep. 2024 – Present*

• Aim to develop a hybrid modeling framework integrating process-based models, knowledge-guided machine learning methods, and Earth observation data.

• Compile, synthesize, and standardize pasture productivity, management, soil, and climate datasets from literature, field measurements, and public data sources.

• Prepare and parameterize datasets for process-based modeling to support model calibration and subsequent machine learning analyses.

Responses of greenhouse gas fluxes to altered precipitation, land management, and cattle grazing

Research Assistant *Sep 2024 – Present*

• Conduct fieldwork in a Florida Long-Term Agroecosystem Research (LTAR) station, focusing on soil gas flux measurements and soil sampling.

• Perform laboratory analyses of soil and gas samples.

• Support the coordination of routine field campaigns, including logistics, sample handling, and communication among research team members.

Root Responses to Global Change Factors

Research Assistant *Feb 2025 – Present*

• Conduct root sampling under different experimental treatments in LTAR station.

• Wash, sort, and analyze root samples to quantify root morphology and trait responses.

Zhejiang University

Modeling the Relationship between Tree Species Diversity in Forest Plantations and Carbon Sequestration Capacity under Projected Climate Change Scenarios

Project Leader, Advised by Dr. Shuang Liang and Dr. Weile Chen *May 2023 – May 2024*

• Synthesized regional-scale plant trait data from literature and public datasets.

• Parameterized soil and climate inputs in the landscape forest model *LANDIS-II* and analyzed simulation outputs.

Integrated Field Course of Biology

Group Leader, supervised by Dr. Jun Chen and Dr. Yunpeng Zhao *Jun 2022 – Jul 2022*

• Conducted ecological field investigations including vegetation surveys and environmental measurements.

• Integrated field data for ecological analysis and interpretation.

PBLICATIONS

- Fu, F., & Qiu, J. Coupling Process-Based Modeling and Machine Learning to Predict Grassland Productivity and Environmental Responses in Florida. (in prep)
- Cai, L., Cui, W., Fu, F., Chen, W., Liang, S. Mixed-species plantations sustain long-term, climate-resilient carbon benefits over monocultures in subtropical China. (*Manuscript in submission*)

Leadership Activities

***Lab manage*r** in Jiangxiao Qiu's Lab *Feb 2025 – Apr 2025*

- Coordinated monthly fieldwork activities, including communication with LTAR groups to confirm field sites, availability, and logistics.
- Presented the team's progress and field plans and helped organize participation in a way that was reasonable and respectful of members' schedules.
- Managed field equipment and organized collected samples with careful attention to accuracy and responsibility.
- Built communication and coordination skills through negotiation, scheduling, and cross-group collaboration.

Project Research leader *Mar 2025 – present,* **University of Florida**

- Took initiative in leading my own research project by actively addressing conceptual and modeling challenges.
- Reached out to experts for guidance and discussion, which helped advance the project and build connections across related fields.

Department Badminton Team Captain *May 2022 – Jun 2024,* **Zhejiang University**

- leading the team to record-breaking results for two consecutive years and a fourth-place finish in the university-wide competition.
- Fostered an inclusive and motivating team environment by organizing training, supporting teammates of different skill levels, and encouraging broad participation in competition.

SKILLS & INTERESTS

Skills:

- Programming & Analysis: R, SAS, Python
- Modeling: Landis-II, APSIM
- GIS & Remote Sensing: ArcGIS, Google Earth Engine (GEE), ENVI 6.0
- Field & Lab Methods

Interests:

- Global Change Ecology
- Ecological Modeling
- Landscape Ecology
- Remote Sensing